DO NOW — RECALL

Insertion sort: trace and implement

Name: _____ Class: _____ Date: _____

1. What is [5,2,4] after one complete ascending bubble-sort pass?

2. What is the value of i after i++ when i was 2?

3. Is a stack FIFO or LIFO?

4. What change makes bubble sort terminate early on sorted data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

[2,4,5].

2. Model answer

3.

3. Model answer

LIFO: last in, first out.

4. Model answer

Track whether any swap occurred; if none occurred during a complete pass, stop.

Vocabulary — know these terms

key — Value being inserted into the sorted prefix.

sorted prefix — Initial segment already placed in order.

shift — Move a value one index to create space.

in-place — Uses little additional data storage.

stable — Equal elements can preserve relative order.

NEW LEARNING

Read, annotate and explain

One key at a time

The prefix to the left of i is already sorted; take a[i] as the key.

Make room

Shift larger values in the sorted prefix one position to the right.

Insert the key

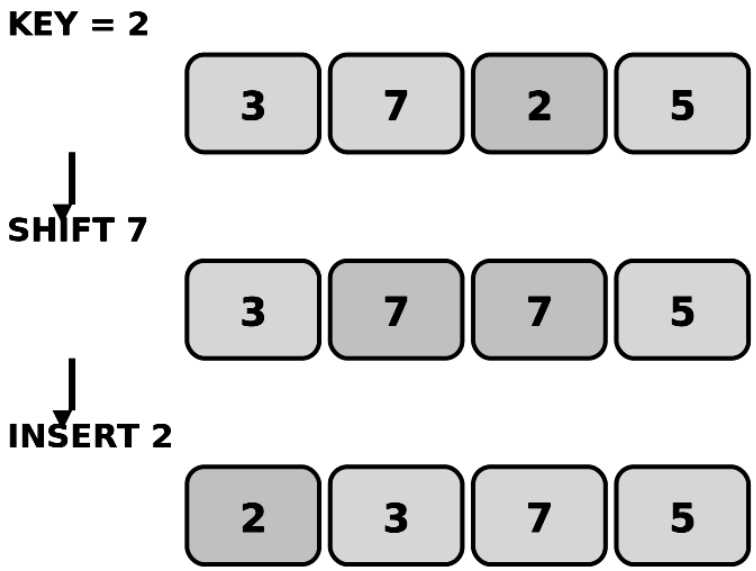
Place it at j+1; the sorted prefix grows by one element.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read the code; predict its effect



Worked algorithm states: KEY = 2 3,7,2,5; SHIFT 7 3,7,7,5; INSERT 2 2,3,7,5

I DO — WORKED EXAMPLE

Follow the worked method

Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].

Predict first

Identify the shifts needed.

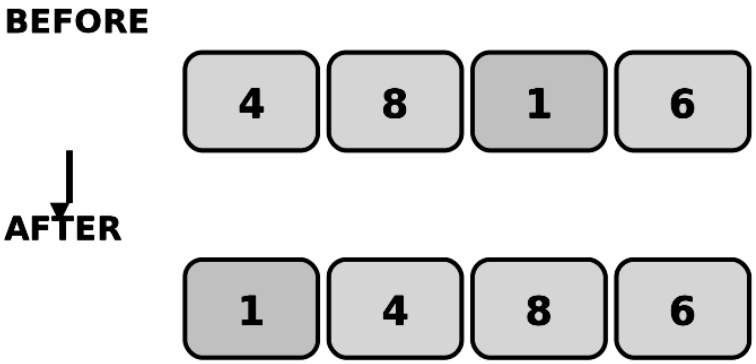
Model solution

Shift 8 then 4 to the right and insert 1 at index 0 → [1,4,8,6].

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].



Worked algorithm states: BEFORE 4,8,1,6; AFTER 1,4,8,6

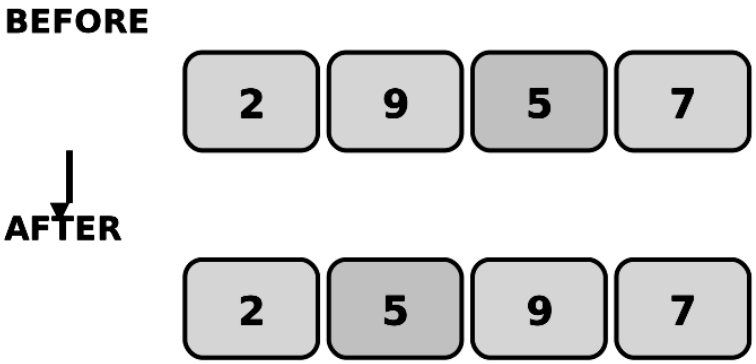
WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert key 5 into the sorted prefix [2,9] of [2,9,5,7].



Worked algorithm states: BEFORE 2,9,5,7; AFTER 2,5,9,7

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Insert key 3 into [2,6,3,8], assuming [2,6] is sorted.

Question 2

Trace insertion sort fully on [5,1,4].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why insertion sort is efficient on almost-sorted input.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Insertion sort always swaps each element with its neighbour.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.